软件分析

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Static Program Analysis

Static Analysis for Security

Nanjing University

Tian Tan

2021

Security

Achieving some goals in the presence of adversaries

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Physical World

- Goals
  - Personal safety
  - Property safety
  - …
- Adversaries
  - Thieves
  - Criminals
  - …

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Physical World

- Goals
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Cyber World

- Goals
  - Dependability
  - Data safety
  - …
- Adversaries
  - Crackers
  - Cyber attackers
  - …

Computer Security

Becomes increasingly important nowadays

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Top 10 Web Application Security Risks [1]

1. **Injection.** Injection flaws, such as SQL, NoSQL, OS, and l
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2. **Broken Authentication**. Application functions related to a
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3. **Sensitive Data Exposure**. Many web applications and AP

Causes of exploited vulnerabilities in 2013-2019[2]

- Injection errors (No. 1), 11821, 4.6/day
- Information leaks (No. 4), 5086, 2.0/day

[1] The Open Web Application Security Project, https://owasp.org/
[2] National Vulnerability Database, https://nvd.nist.gov/

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1. Information Flow Security
2. Confidentiality and Integrity
3. Explicit Flows and Covert Channels
4. Taint Analysis

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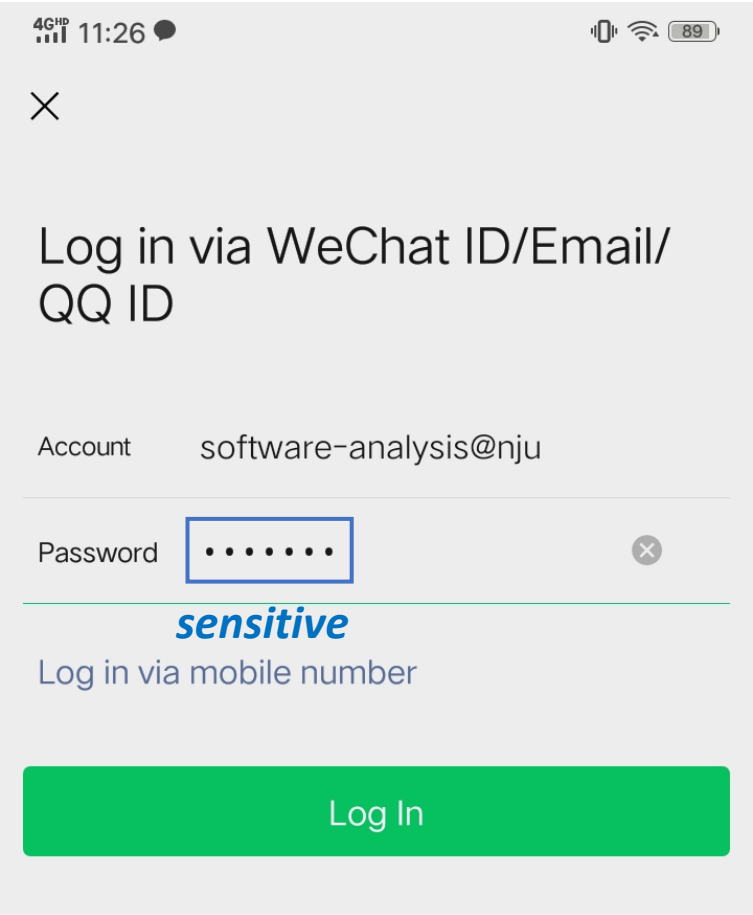
Information Flow: An Example



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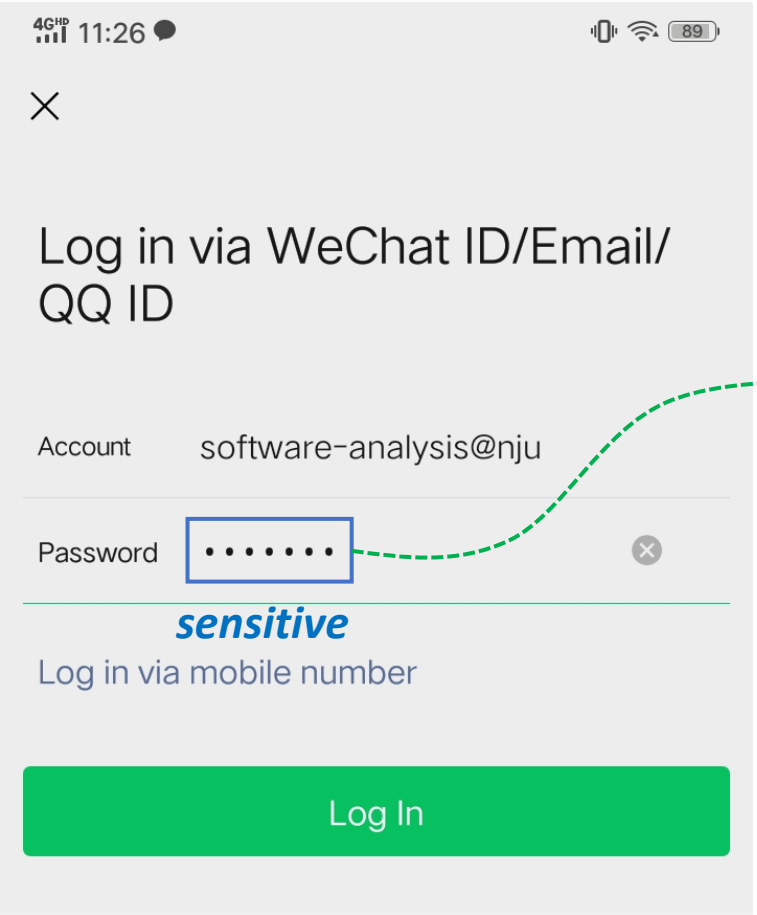

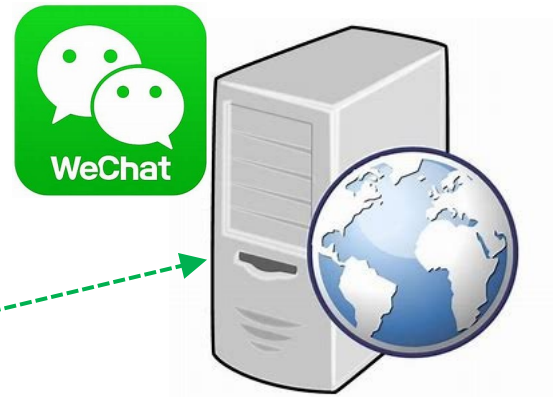

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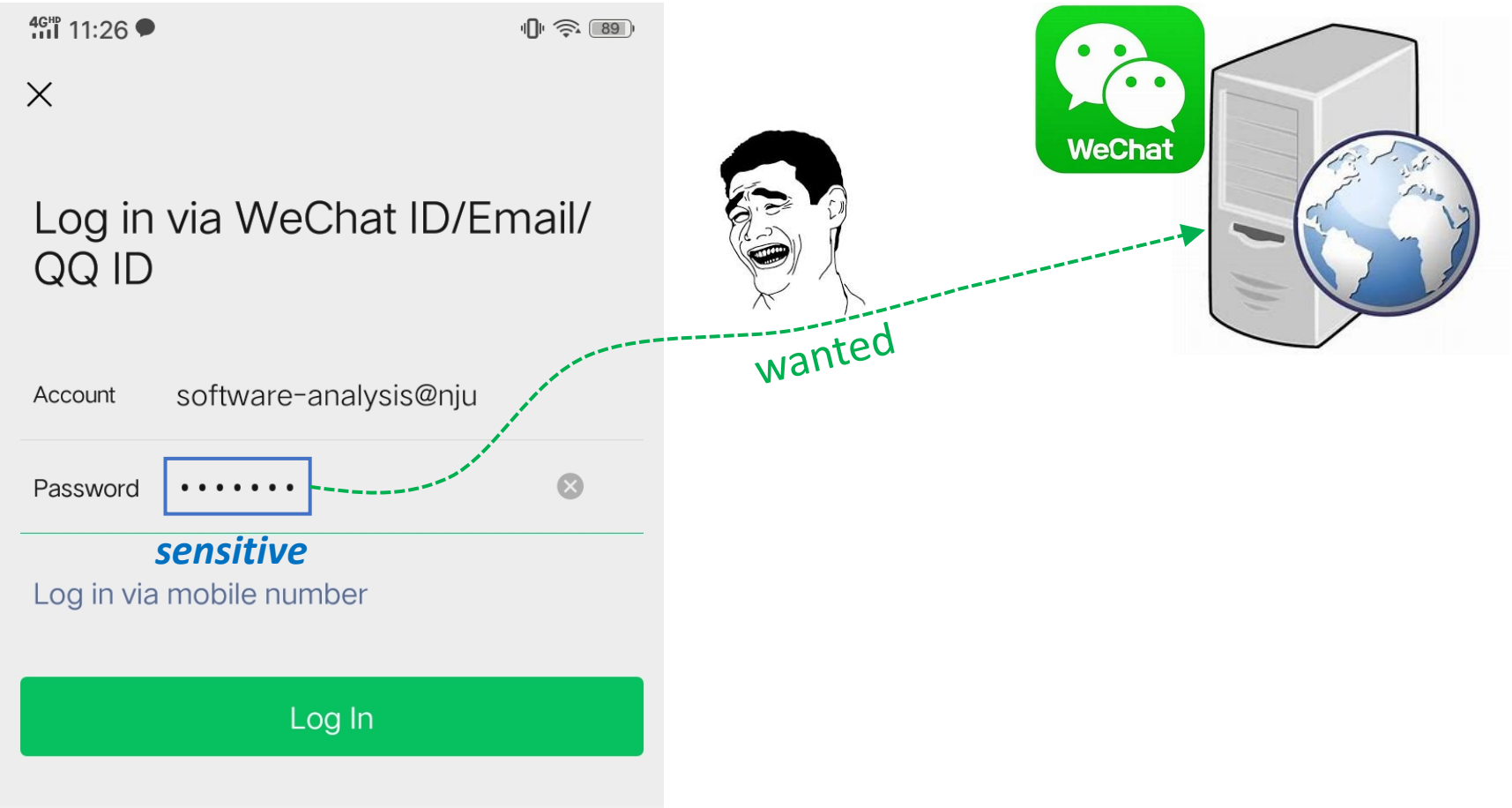

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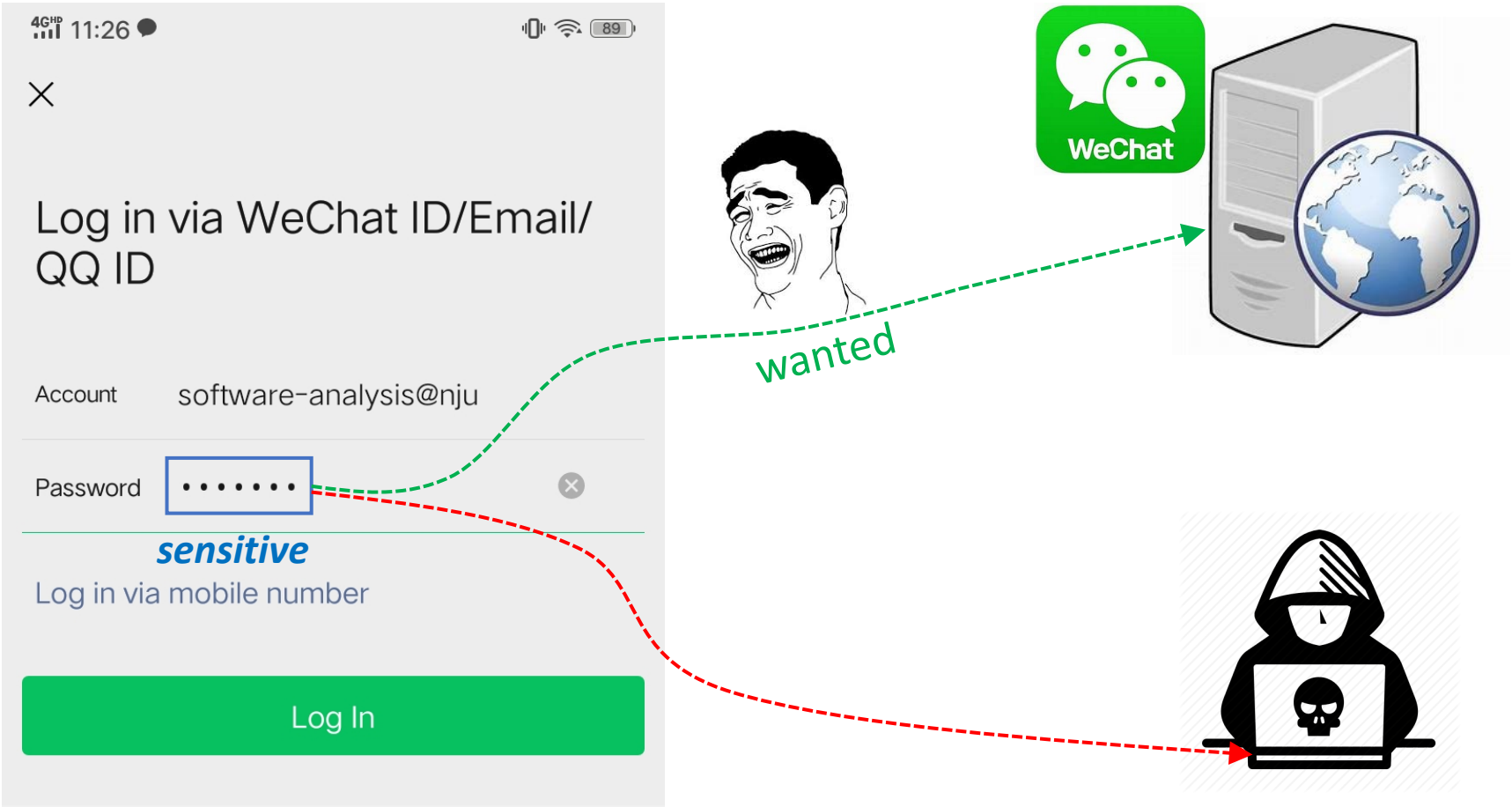

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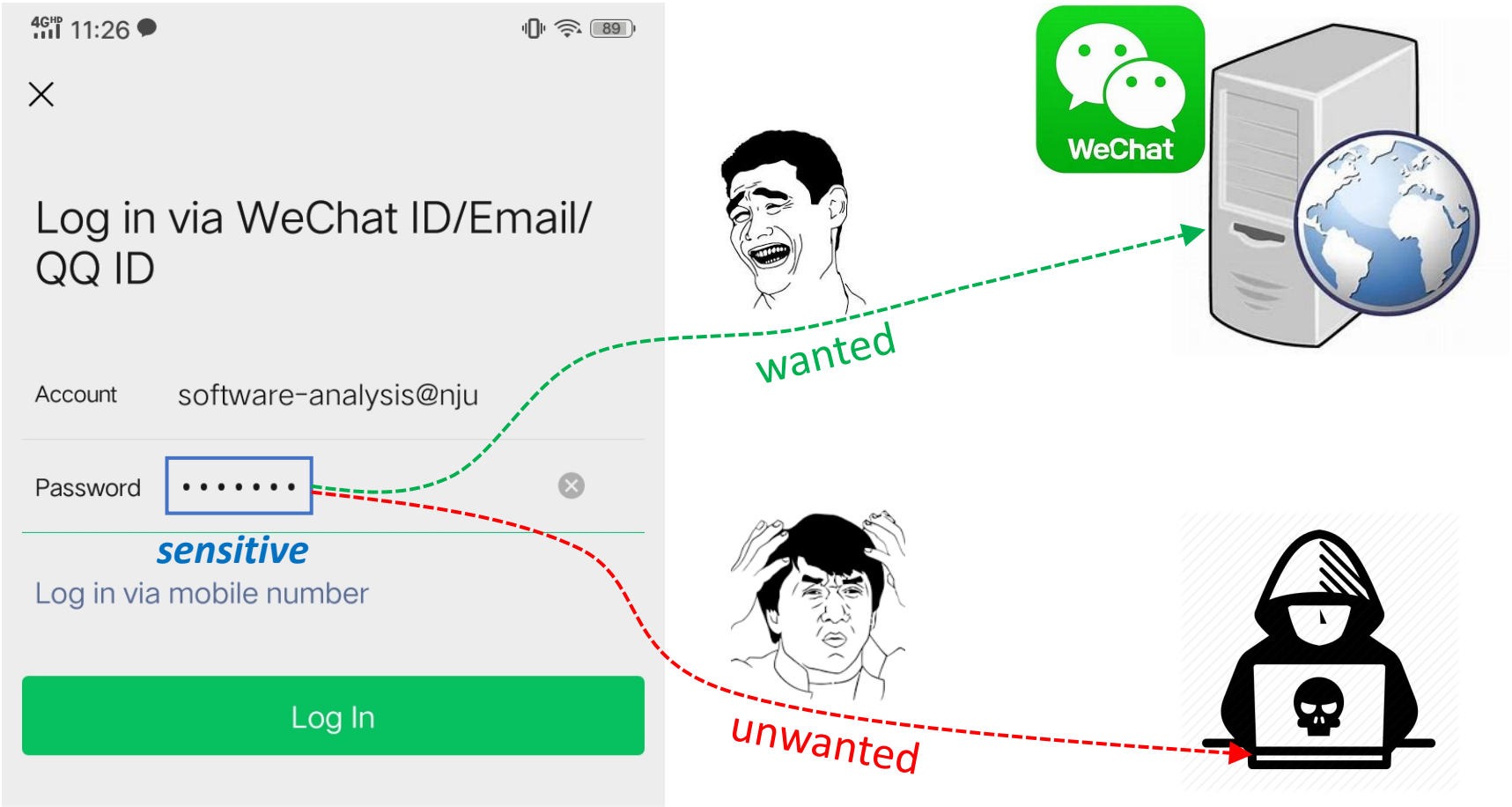
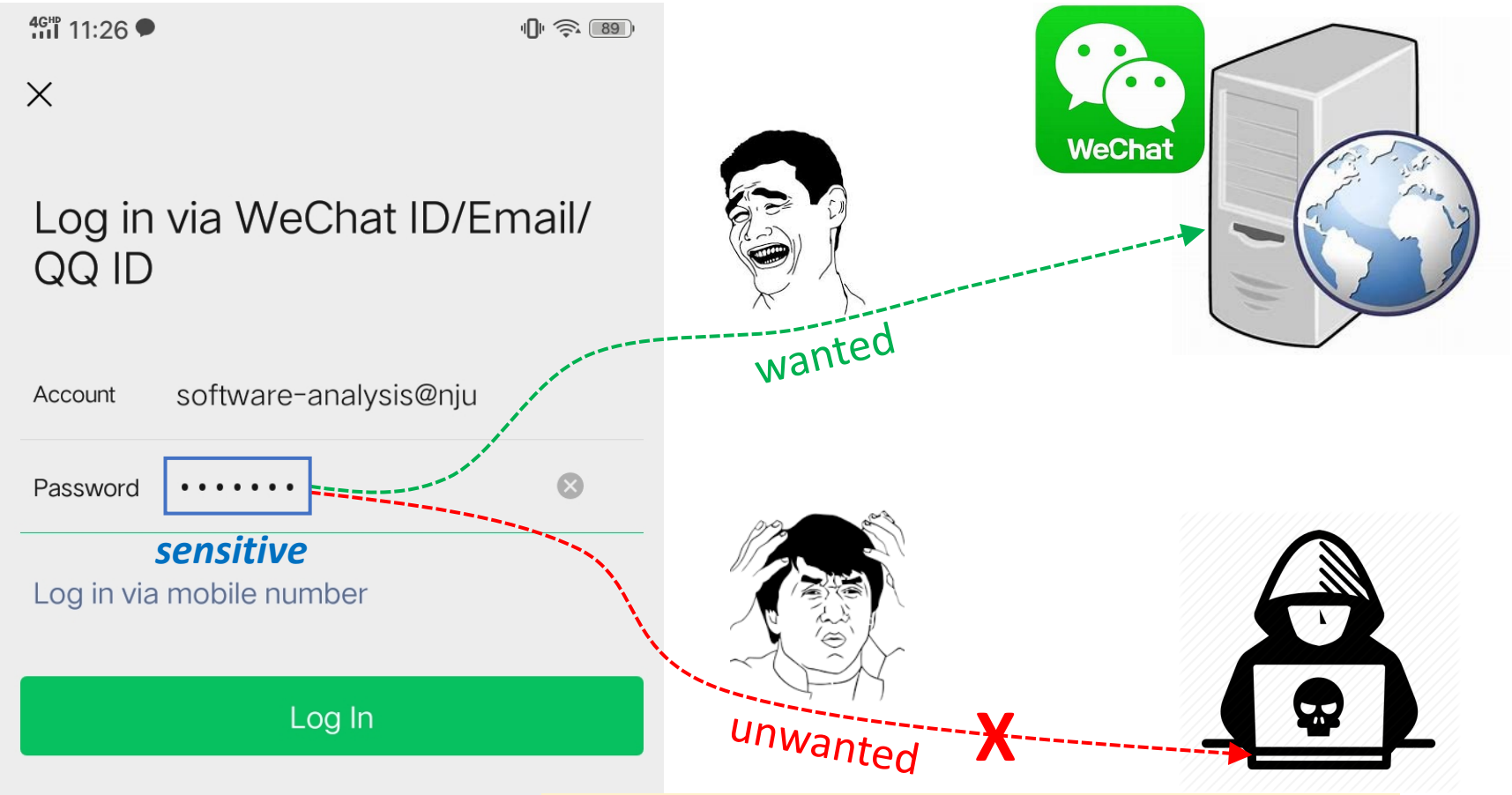


Information Flow Security: Motivation



Prevent unwanted information flow
Protect information security

Access Control vs. Information Flow Security

- Access control (a standard way to protect sensitive data)
  - Checks if the program has the rights/permissions to access certain information
  - Concerns how information is **accessed**

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仅仅有访问控制是不够的

- Information flow security (end-to-end)
  - Tracks how information flows through the program to make sure that the program handles the information securely
  - Concerns how information is **propagated**

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"*A practical system needs both **access** and **flow control** to satisfy all security requirements.*"
— D. Denning, 1976

Information Flow*

- Information flow: if the information in variable x is transferred to variable y, then there is information flow $x \rightarrow y$

*Dorothy E. Denning and Peter J. Denning, "*Certification of Programs for Secure Information Flow*". CACM 1977.

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- Examples

```
y = x;
```

```
a = x;
b.f = a;
y = b.f;
```

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Information Flow*

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- Examples

Looks like pointer analysis?
You will see …

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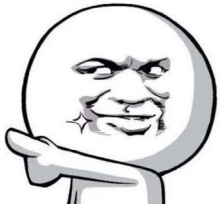

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Information Flow Security

Connects information flow to security

- Classifies program variables into different security levels
- Specifies permissible flows between these levels, i.e., information flow policy

Security Levels (Classes)

- The most basic model is two-level policy, i.e., a variable is classified into one of two security levels:
  1. H, meaning *high* security, secret information
  2. L, meaning *low* security, public observable information

```
h = getPassword(); // h is high security
broadcast(l);      // l is low security
```

Security Levels (Classes)

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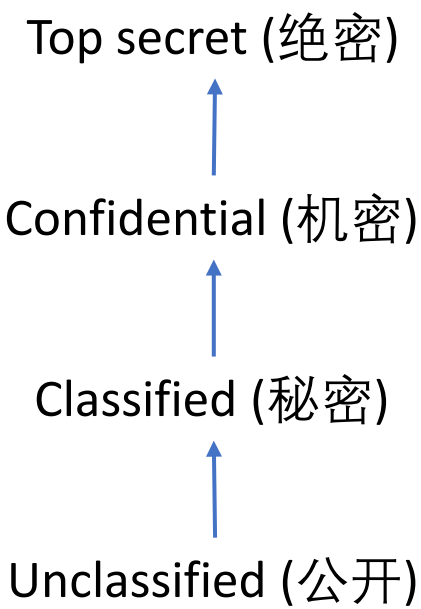
- Security levels can be modeled as *lattice*∗
  - L ≤ H

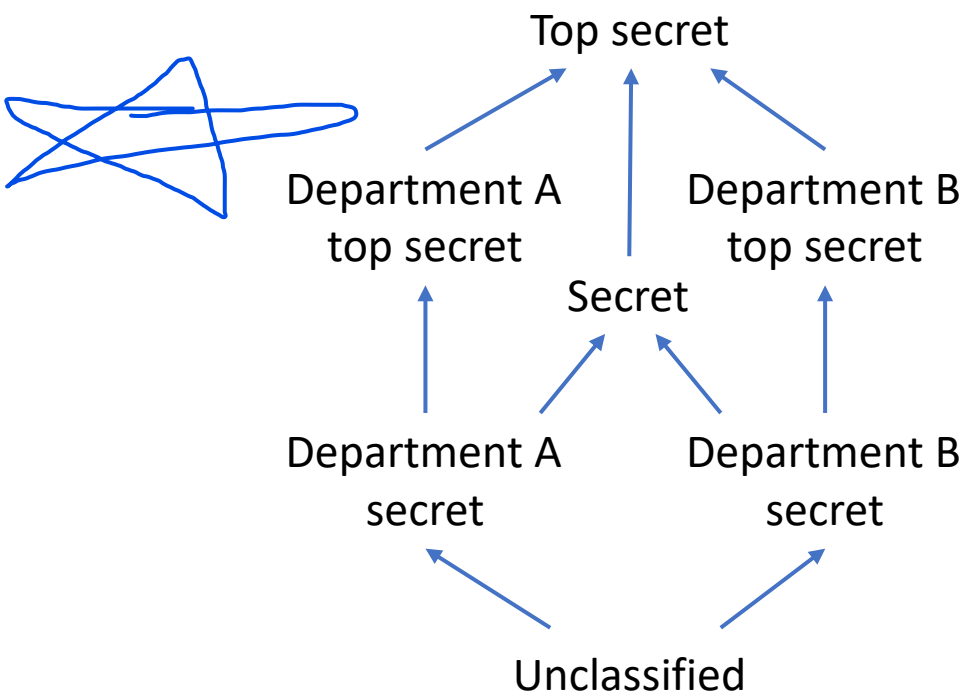
∗Dorothy E. Denning, "*A Lattice Model of Secure Information Flow*". CACM 1976.

More Complicated Security Levels

- China classification



- A (possible) business classification



Information Flow Policy

- Restricts how information flows between different security levels

Information Flow Policy

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- Noninterference policy*
  - Requires the information of high variables have no effect on (i.e., should not interfere with) the information of low variables
  - Intuitively, you should not be able to conclude anything about high information by observing low variables

*J. A. Goguen and J. Meseguer, "*Security policies and security models*". S&P 1982.

Noninterference

- Requires the information of high variables have no effect on (i.e., should not interfere with) the information of low variables
- ✓ $x_H = y_H$
- ✓ $x_L = y_L$

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? • $x_L = y_H$

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✗ • $x_L = y_H$

✓ • $x_H = y_L$

? • $x_L = y_L + z_H$

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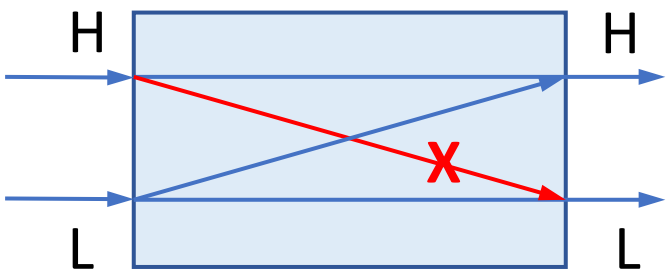

Ensures that information flows only upwards in the lattice

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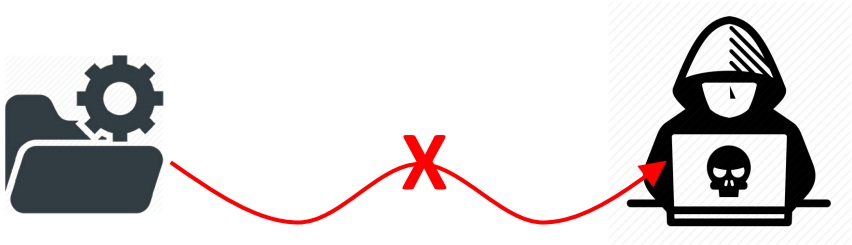
- Confidentiality 机密性
  - Prevent secret information from being leaked



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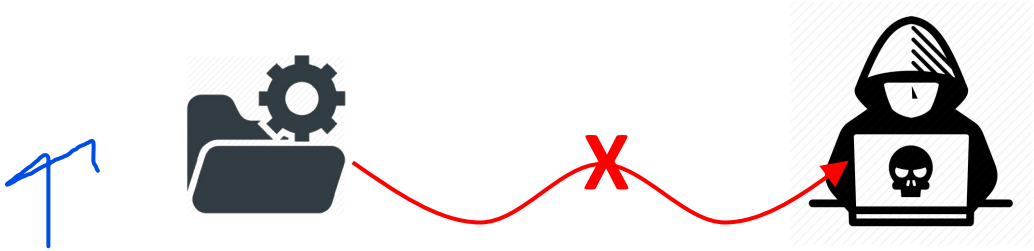
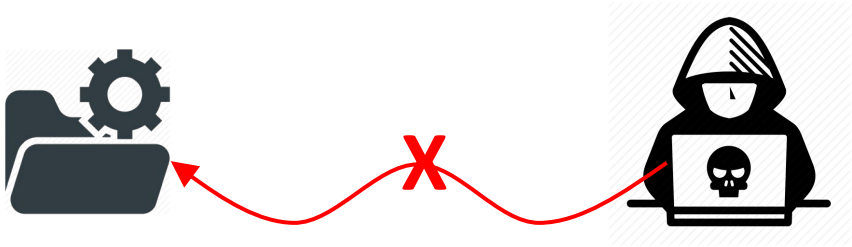


Information flow security from another perspective

- Integrity 完整性
  - Prevent untrusted information from corrupting (trusted) critical information



Integrity

- Prevent untrusted information from corrupting (trusted) critical information[1]

```
x = readInput(); // untrusted
cmd = "..." + x;
execute(cmd); // critical (trusted)
```

1. Ken Biba, "*Integrity Considerations for Secure Computer Systems*". Technical Report, ESD-TR-76-372, USAF Electronic Systems Division, Bed-ford, MA, 1977.

Integrity

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- Injection errors (#1 cause of vulnerabilities in 2013-2019[2])
  - Command injection
  - SQL injection
  - XSS attacks
  - …

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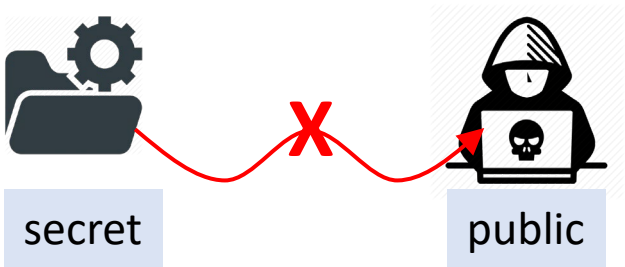
2. National Vulnerability Database, https://nvd.nist.gov/

Confidentiality and Integrity

对偶

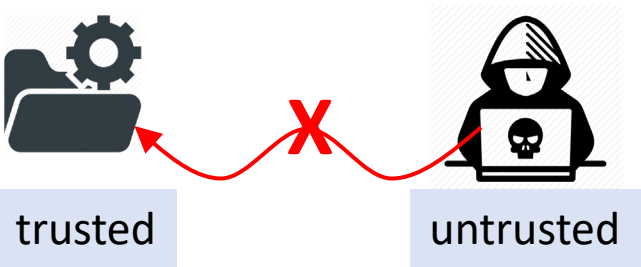
Confidentiality **Duals** **Integrity**

- Security classification
  - Secret (**H**igh secret) **H**
  - Public (**L**ow secret) **L**



- Read protection

- Security classification
  - Trusted (**H**igh integrity) **L**
  - Untrusted (**L**ow integrity) **H**



- Write protection

Integrity, Broad Definition

- To ensure the correctness, completeness, and consistency of data

- Correctness
  - E.g., for information flow integrity, the (trusted) critical data should not be corrupted by untrusted data
- Completeness
  - E.g., a database system should store all data completely
- Consistency
  - E.g., a file transfer system should ensure that the file contents of both ends (sender and receiver) are identical

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How Does Information Flow

- $x_H = y_H$
- $x_L = y_H$
- $x_L = y_L + z_H$

We have seen how information flows through direct copying. This is called explicit flow.

How Does Information Flow

- $x_H = y_H$
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We have seen how information flows through direct copying. This is called explicit flow.

Is this the only way of information flow?

Does Secret Information Leak?

```
secretH = getSecret();
if (secretH < 0)
    publikL = 1;
else
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- This kind of information flow is called **implicit flow**, which may arise when the control flow is affected by secret information.

- Any differences in side effects under secret control encode information about the control, which may be publicly observable and leak secret information.

Implicit Flows

Are there *any other* kinds of information flows?

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Does Secret Information Leak?

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while (secret_H < 0) { … };
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```
if (secret_H < 0)
  for (int i = 0; i < 1000000; ++i) { … };
```

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if (secretH < 0)
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if (secretH < 0)
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?

```
int saH[] = getSecretArray();
saH[secretH] = 0;
```

Does Secret Information Leak?

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int saH[] = getSecretArray();
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Leak, exception may reveal that secret is negative

Covert/Hidden Channels

- Mechanisms for signalling information through a computing system are known as ***channels***.
- Channels that exploit a mechanism whose primary purpose is not information transfer are called ***covert channels****.

隐藏信道

*Butler W. Lampson, "*A Note on the Confinement Problem*". CACM 1973.

Covert/Hidden Channels

- Mechanisms for signalling information through a computing system are known as ***channels***.
- Channels that exploit a mechanism whose primary purpose is not information transfer are called ***covert channels****.

- Implicit flows `if (secret_H < 0) p_L = 1; else p_L = 0;`

  signal information through the control structure of a program

- Termination channels `while (secret_H < 0) { … };`

  signal information through the (non)termination of a computation

- Timing channels `if (secret_H < 0) for (…) { … };`

  signal information through the computation time

- Exceptions `if (secret_H < 0) throw new Exception("…");`

  signal information through the exceptions

- …

*Butler W. Lampson, "*A Note on the Confinement Problem*". CACM 1973.

Explicit Flows and Covert Channels

- Explicit flows generally carry more information than covert channels, so we focus on explicit flows

```
int secret_H = getSecret();
int publik_L = secret_H;
```

Explicit flow:
transmits 32 bits of information

```
int secret_H = getSecret();
if (secret_H % 2 == 0)
    publik_L = 1;
else
    publik_L = 0;
```

Implicit flow:
transmits 1 bit of information

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Implicit flow:
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How to prevent unwanted information flows, i.e., enforce information flow policies?

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Taint Analysis

- Taint analysis is the most common information flow analysis. It classifies program data into two kinds:
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  - Other data, called untainted data

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  - Other data, called untainted data
- Sources of tainted data is called **sources**. In practice, tainted data usually come from the return values of some methods (regarded as sources).
- Taint analysis tracks how tainted data flow through the program and observes if they can flow to locations of interest (called **sinks**). In practice, sinks are usually some sensitive methods.

Taint Analysis: Two Applications

- Confidentiality
  - Source: source of secret data
  - Sink: leakage
  - Information leaks

```
x = getPassword(); // source
y = x;
log(y); // sink
```

- Integrity
  - Source: source of untrusted data
  - Sink: critical computation
  - Injection errors

```
x = readInput(); // source
cmd = "..." + x;
execute(cmd); // sink
```

Taint analysis can detect **both** unwanted information flows

Taint Analysis

- “Can tainted data flow to a sink?”

Taint Analysis

- “Can tainted data flow to a sink?”

Or, in another way

- “Which tainted data a pointer (at a sink) can point to?”

Taint and Pointer Analysis, Together*

The essence of taint analysis/pointer analysis is to track how tainted data/abstract objects flow through the program

- Treats tainted data as (artificial) objects
- Treats sources as allocation sites (of tainted data)
- Leverages pointer analysis to propagate tainted data

*Neville Grech and Yannis Smaragdakis, "*P/Taint: Unified Points-to and Taint Analysis*". OOPSLA 2017.

Domains and Notations

Variables:	$x, y \in \mathrm{V}$
Fields:	$f, g \in \mathrm{F}$
Objects:	$o_i, o_j \in \mathrm{O}$
Tainted data:	$t_i, t_j \in \mathrm{T} \subset \mathrm{O}$
Instance fields:	$o_i.f, o_j.g \in \mathrm{O} \times \mathrm{F}$
Pointers:	$\mathrm{Pointer} = \mathrm{V} \cup (\mathrm{O} \times \mathrm{F})$
Points-to relations:	$pt : \mathrm{Pointer} \rightarrow \mathcal{P}(\mathrm{O})$

- t_i denotes the tainted data from call site i
- $\mathcal{P}(\mathrm{O})$ denotes the powerset of O
- $pt(p)$ denotes the points-to set of p

Domains and Notations

Variables: $x, y \in \mathrm{V}$

Fields: $f, g \in \mathrm{F}$

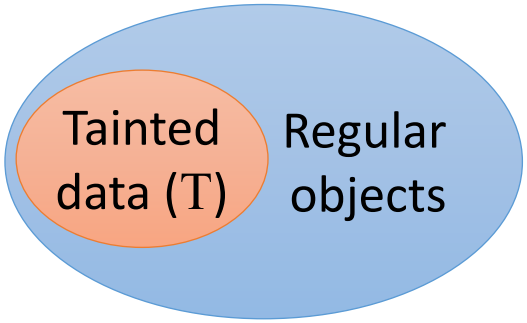
Objects: $o_i, o_j \in \mathrm{O}$

Tainted data: $t_i, t_j \in \mathrm{T} \subset \mathrm{O}$

Instance fields: $o_i.f, o_j.g \in \mathrm{O} \times \mathrm{F}$

Pointers: $\mathrm{Pointer} = \mathrm{V} \cup (\mathrm{O} \times \mathrm{F})$

Points-to relations: $pt : \mathrm{Pointer} \rightarrow \mathcal{P}(\mathrm{O})$



All objects (O)

- t_i denotes the tainted data from call site i
- $\mathcal{P}(\mathrm{O})$ denotes the powerset of O
- $pt(p)$ denotes the points-to set of p

Taint Analysis: Inputs & Outputs

- Inputs
  - $Sources$: a set of source methods (the calls to these methods return tainted data)
  - $Sinks$: a set of sink methods with sensitive arguments (that tainted data flow to the arguments of these methods violates security polices)

- Outputs
  - $TaintFlows$: a set of tuples of source and sink calls

    E.g., $\langle i, j, k \rangle \in TaintFlows$ denotes that the tainted data from call site i (which calls a source method) may flow to the k-th argument of call site j (which calls a sink method)

Rules: Call

- Handles sources (generates tainted data)

Kind	Statement	Rule
Call	`l: r = x.k(a1,…,an)`	$\frac{l \rightarrow m \in CG \quad m \in \mathit{Sources}}{t_l \in pt(r)}$

Rules (Same as Pointer Analysis)

Kind	Statement	Rule
New	`i: x = new T()`	$\overline{o_i \in pt(x)}$
Assign	`x = y`	$\frac{o_i \in pt(y)}{o_i \in pt(x)}$
Store	`x.f = y`	$\frac{o_i \in pt(x),\ o_j \in pt(y)}{o_j \in pt(o_i.f)}$
Load	`y = x.f`	$\frac{o_i \in pt(x),\ o_j \in pt(o_i.f)}{o_j \in pt(y)}$
Call	`l: r = x.k(a1,…,an)`	$\frac{\begin{array}{c} o_i \in pt(x),\ m = \text{Dispatch}(o_i, \mathrm{k}) \\ o_u \in pt(aj), 1 \le j \le n \\ o_v \in pt(m_{ret}) \end{array}}{\begin{array}{c} o_i \in pt(m_{this}) \\ o_u \in pt(m_{pj}), 1 \le j \le n \\ o_v \in pt(r) \end{array}}$

Propagate objects and tainted data

Rules: Call

- Handles sources (generates tainted data)

Kind	Statement	Rule
Call	`l: r = x.k(a1,…,an)`	$\frac{l \rightarrow m \in CG \quad m \in Sources}{t_l \in pt(r)}$

- Handles sinks (generates taint flow information)

Kind	Statement	Rule
Call	`l: r = x.k(a1,…,an)`	$\frac{l \rightarrow m \in CG \quad \langle m, i \rangle \in Sinks \quad t_j \in pt(ai)}{\langle j, l, i \rangle \in TaintFlows}$

Taint Analysis: An Example

```
1  void main() {
2    A x = new A();
3    String pw = getPassword();
4    A y = x;
5    x.f = pw;
6    String s = y.f;
7    log(s);
8  }
9  String getPassword() {
10   …
11   return new String(…);
12 }
13 class A {
14   String f;
15 }
```

Sources: { getPassword() }

Sinks: { ⟨log(String), 0⟩ }

Taint Analysis: An Example

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1  void main() {
2    A x = new A();
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15 }
```

Sources: { getPassword() }

Sinks: { ⟨log(String), 0⟩ }

Variable	Object
x	o_2

Taint Analysis: An Example

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1  void main() {
2    A x = new A();
3    String pw = getPassword();
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7    log(s);
8  }
9  String getPassword() {
10   …
11   return new String(…);
12 }
13 class A {
14   String f;
15 }
```

Sources: { getPassword() }

Sinks: { ⟨log(String), 0⟩ }

Variable	Object
x	o_2
pw	o_{11}

Taint Analysis: An Example

```
1  void main() {
2    A x = new A();
3    String pw = getPassword();
4    A y = x;
5    x.f = pw;
6    String s = y.f;
7    log(s);
8  }
9  String getPassword() {
10   …
11   return new String(…);
12 }
```

Sources: { getPassword() }

Sinks: { ⟨log(String), 0⟩ }

Variable	Object
x	o_2
pw	o_{11}, t_3

Kind	Statement	Rule
Call	`l: r = x.k(a1,…,an)`	$\frac{l \rightarrow m \in CG \quad m \in Sources}{t_l \in pt(r)}$

Taint Analysis: An Example

```
1  void main() {
2    A x = new A();
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10   …
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12 }
13 class A {
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15 }
```

Sources: { getPassword() }

Sinks: { ⟨log(String), 0⟩ }

Variable	Object
x	o_2
pw	o_{11}, t_3
y	o_2

Taint Analysis: An Example

```
1  void main() {
2    A x = new A();
3    String pw = getPassword();
4    A y = x;
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```

Sources: { getPassword() }

Sinks: { ⟨log(String), 0⟩ }

Variable	Object
x	o_2
pw	o_{11}, t_3
y	o_2
Field	**Object**
o_2.f	o_{11}, t_3

Taint Analysis: An Example

```
1  void main() {
2    A x = new A();
3    String pw = getPassword();
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Sources: { getPassword() }

Sinks: { ⟨log(String), 0⟩ }

Variable	Object
x	o_2
pw	o_{11}, t_3
y	o_2
s	o_{11}, t_3
Field	**Object**
o_2.f	o_{11}, t_3

Taint Analysis: An Example

```
1  void main() {
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7    log(s);
8  }
9  String getPassword() {
10   …
11   return new String(…);
12 }
```

Sources: { getPassword() }

Sinks: { ⟨log(String), 0⟩ }

TaintFlows: { ⟨3, 7, 0⟩ }

Variable	Object
x	o_2
pw	o_{11}, t_3
y	o_2
s	o_{11}, t_3
Field	**Object**
o_2.f	o_{11}, t_3

Kind	Statement	Rule
Call	l: r = x.k(a1,…,an)	$\dfrac{l \rightarrow m \in CG \quad \langle m, i\rangle \in Sinks \quad t_j \in pt(ai)}{\langle j, l, i\rangle \in TaintFlows}$

Taint Analysis: An Example

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1  void main() {
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```

Sources: { getPassword() }

Sinks: { ⟨log(String), 0⟩ }

TaintFlows: { ⟨3, 7, 0⟩ }

Variable	Object
x	o_2
pw	o_{11}, t_3
y	o_2
s	o_{11}, t_3
Field	**Object**
o_2.f	o_{11}, t_3

The X You Need To Understand in This Lecture

- Concept of information flow security
- Confidentiality and integrity
- Explicit flows and covert channels
- Use taint analysis to detect unwanted information flow